

Name: _____

Date: _____

RATIO, (SIMPLIFYING RATIOS)

LO: I can simplify ratios by dividing both sides by their highest common factor.

Simplifying ratios

To **simplify a ratio**, divide all parts by their **highest common factor (HCF)**. A ratio is in its simplest form when the parts share no common factor other than 1.

Remember: Ratios can include units. Convert to the same unit first, then simplify.

Quick examples

●	12 : 18 = 2 : 3	divide both by HCF 6
●	1.5 : 2.5 = 3 : 5	multiply by 2 first to remove decimals
●	40 cm : 1 m = 40 : 100 = 2 : 5	convert to same unit
●	12 : 18 = 6 : 9	not fully simplified
●	40 cm : 1 m = 40 : 1	units not converted

Key idea

Divide every part of the ratio by the highest common factor to simplify.

$$a : b = \frac{a}{\text{HCF}} : \frac{b}{\text{HCF}}$$

Keep dividing until no common factor remains

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - whole numbers

Simplify 24 : 36

Find HCF of 24 and 36 = 12

$$24 \div 12 = 2$$

$$36 \div 12 = 3$$

2 : 3

Always find the largest factor that divides both numbers.

Example 2 - three-part ratio

Simplify 10 : 15 : 25

Find HCF of 10, 15, and 25 = 5

Divide each: $10 \div 5 = 2$, $15 \div 5 = 3$,

$$25 \div 5 = 5$$

Simplified ratio: 2 : 3 : 5

2 : 3 : 5

The same method works for three-part ratios.

Example 3 - decimals

Simplify 0.6 : 1.4

Multiply both by 10: 6 : 14

HCF of 6 and 14 = 2

$$6 \div 2 = 3, 14 \div 2 = 7$$

3 : 7

Remove decimals first by multiplying by 10, 100, etc.

Example 4 - different units

Simplify 500 g : 2 kg

Convert to same unit: 500 g : 2000

g

HCF of 500 and 2000 = 500

$$500 \div 500 = 1, 2000 \div 500 = 4$$

1 : 4

Always convert to the same unit before simplifying.

YOUR TURN – PRACTICE (deliberate practice)

1. Simple whole number ratios

Simplify each ratio fully.

- a) 8 : 12
- b) 15 : 25
- c) 18 : 24
- d) 30 : 45

2. Larger numbers

Find the HCF and simplify.

- a) 48 : 64
- b) 72 : 108
- c) 56 : 84
- d) 90 : 126

3. Decimals and fractions

Simplify these ratios.

- a) 0.4 : 0.6
- b) 1.2 : 1.8
- c) $\frac{1}{2} : \frac{3}{4}$
- d) 2.5 : 3.5

4. Different units

Convert to the same unit, then simplify.

- a) 50 cm : 2 m
- b) 300 g : 1.5 kg
- c) 20 min : 1 hour
- d) 750 ml : 2 litres

5. Mixed simplifying

Simplify each ratio fully.

- a) 36 : 48 : 60
- b) 0.8 : 1.2 : 2
- c) $\frac{2}{3} : \frac{5}{6}$
- d) 45 cm : 1.5 m : 2.25 m
- e) 125 : 200 : 350

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) 12 : 18 simplifies to 2 : 3
(divide by 6) ☐

Correct answer: _____

Reason: _____

b) 12 : 18 simplifies to 6 : 9 and
that is the simplest form ☐

Correct answer: _____

Reason: _____

c) To simplify a ratio with
decimals, multiply to make
whole numbers first ☐

Correct answer: _____

Reason: _____

d) 40 cm : 1 m = 40 : 1 because we
just drop the units ☐

Correct answer: _____

Reason: _____

e) A ratio is fully simplified when
the HCF of all parts is 1 ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A recipe uses **250 g of flour**, **150 g of sugar**, and **100 g of butter**. Write the ratio flour : sugar : butter in simplest form. If you only have **200 g of butter**, how much flour and sugar do you need?

Expression: _____

Simplified: _____

b) The angles in a quadrilateral are in the ratio **1 : 2 : 3 : 4**. Find each angle. The smallest angle is increased by **10 degrees**. If the shape remains a quadrilateral, what must the largest angle become?

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

